



Final Project Report

“Analysis of Customer Purchase Behavior and Marketing Campaign Effectiveness Using Retail Transaction Data”

Capstone Project

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1. Executive Summary

1.1. Overview

Retail businesses invest heavily in promotional campaigns, yet most lack the analytical tools to determine whether those campaigns generate genuine long-term customer loyalty or simply produce temporary sales spikes that fade once the promotion ends. This project addresses that critical business problem using the Dunnhumby Complete Journey dataset a real-world, two-year loyalty card record covering 2,500 households across transactions, demographics, coupon activity, and 30 distinct marketing campaigns. Applying the CRISP-DM framework, this project's objective was to build a suite of five predictive and analytical models to measure true campaign effectiveness, identify genuinely persuadable customers, and replace the retailer's current one-size-fits-all promotional strategy with a data-driven, personalized marketing approach that maximizes return on investment and builds sustainable customer engagement.

1.2. Key Findings

The analysis uncovered five important findings that shows how customers respond to the promotions.

First, the most impressive discovery was that roughly 75% of customers who received promotional offers were going to buy anyway they did not need a campaign to convert. The Uplift Model (Test Qini: 22.69, Test AUUC: 6.09) confirmed that only the top 25-30% of customers, those with the highest persuadability scores, showed any real behavioral change because of a promotion.

Second, while campaigns did produce a positive effect overall, the impact was smaller and far less consistent than expected. Average customer spending rose by just 2.6% from \$2,771.16 to \$2,843.97 and even that modest gain was uneven: 55.1% of customers spent more during campaigns while 44.9% actually spent less, showing that promotions do not move all customers in the same direction.

Third, and perhaps most importantly for long-term strategy, discounts were found to be the weakest driver of customer loyalty. The Customer Engagement Predictor (Test AUC: 0.86, Recall: 0.90) showed that how much a customer buys (65.4% feature importance) and how recently they shopped (18.6%) are what truly predict sustained engagement — discount dependency ranked last at just 16%, confirming that piling on deeper discounts will not turn a casual shopper into a loyal one.

Fourth, the Coupon Recommendation model (Test AUC: 0.686) revealed that the retailer's best customers those who spend the most and shop most frequently almost never redeem coupons. They buy at full price regardless. Offering them discounts does not change their behavior; it simply reduces the margin earned from customers who were never going to walk away.

Finally, what does drive stronger store performance is product variety. Stores where customers regularly shop across a wider range of categories consistently showed higher visit frequency and greater spend per household. This suggests that expanding what customers engage with across the store is a more powerful and sustainable loyalty lever than any promotional campaign on its own.

1.3. Recommendations

The findings point clearly toward three changes the business should make in how it approaches promotional marketing.

The most immediate change is in **who gets targeted**. Right now, campaigns are being sent broadly but the Uplift Model shows that only a specific group of customers actually changes their behavior because of a promotion. Going forward, the model's persuadability scores should be used as a filter before every campaign goes out, ensuring that promotional offers reach only the customers in the "Very High" uplift segment. This one shift alone would stop roughly 75% of the campaign budget from being spent on customers who were already going to buy, without giving up any of the incremental revenue the campaign was meant to generate.

The second change is in **how discounts are distributed**. Rather than offering the same coupon to every customer, the Coupon Recommendation model makes it possible to match each customer to the right level of incentive: a light 5% discount for customers who engage readily, a moderate 10% for those who need a nudge, and a stronger 20% for customers who are harder to activate. High-value, frequent shoppers would receive little to no discount since they demonstrably buy at full price anyway, while the higher incentives would be reserved for customers where that extra push is actually needed. This approach protects margins without reducing campaign reach.

The third change is a broader strategic shift in **where retention of investment goes**. Since the data consistently shows that what keeps customers coming back is how often they shop and how broadly they engage across product categories—not how deep a discount they received—marketing resources should move toward programs that reward repeat visits, encourage exploration across more product categories, and surface personalized product recommendations. The Product Recommendation model already provides the similarity-based outputs needed to build cross-sell and bundling campaigns that create this kind of deeper engagement, all without relying on margin-eroding discounts to do it.

1.4. Outcome:

Together, these three strategies mark a fundamental change in the way the business thinks about promotional marketing: stepping away from the current approach of sending broad, discount-heavy campaigns to everyone, and moving toward a smarter, more targeted model where decisions are driven by data and every promotional dollar is put to work where it will actually make a difference. The most immediate place that change will show up is in marketing ROI. By concentrating campaign spend on the genuinely persuadable customers identified by the Uplift Model, and routing discounts through the tiered coupon system rather than distributing them uniformly, the business stands to recover a significant portion of the promotional budget that is currently being spent with little to no incremental return.

Beyond immediate cost savings, the shift toward engagement-building strategies rewarding shopping frequency, encouraging category breadth, and personalizing product discovery is designed to produce stronger and more durable customer retention over time. Since purchase volume and recency are the real drivers of loyalty, investing in programs that reinforce those behaviors creates a compounding effect that discount campaigns simply cannot replicate.

Equally important is what this project leaves behind as infrastructure. The before-and-after campaign measurement framework developed here is fully reusable. Every future campaign can now be measured the same way looking at how individual customers actually behaved before and after a promotion, rather than leaning on overall sales numbers that can make a campaign look successful even when the reality at the customer level tells a very different story. That kind of visibility compounds over time. Each campaign becomes an opportunity to learn a little more, tighten the targeting a little further, and make the next round of promotional decisions a little sharper than the last.

In short, the expected outcome is not just a better campaign it is a better way of running campaigns altogether.

2. Business Understanding

2.1. Business Problem Definition:

Every day, retail stores collect massive amounts of data from customer purchases and loyalty programs. But despite sitting on all this information, many companies still struggle to answer a very basic question: Are our marketing campaigns actually working? Too often, retailers rely on a "one-size-fits-all" approach. They blast the exact same coupons and promotional offers to everyone, without taking the time to understand what individual shoppers actually want or how they behave.

This brings us to the core business question our project addresses: **When a store offers a discount, does it genuinely build long-term customer loyalty, or does it just create a quick, temporary spike in sales?**

Solving this problem is critical for the organization. When marketing isn't targeted, companies end up burning through their promotional budgets on offers that don't drive real value. If a retailer cannot accurately measure whether a campaign changed a customer's habits, it directly hurts their bottom line. It makes it impossible to maximize marketing Return on Investment (ROI) and limits their ability to grow Customer Lifetime Value (CLV).

Using the dunnhumby dataset, this project aims to uncover the real story behind the numbers. By combining raw daily transaction data with household demographics, we will figure out if current campaigns are genuinely creating profitable, long-term customer engagement or if shoppers are simply taking the discount and going back to their old habits once the sale ends.

2.2. Objectives:

The primary objective of this data mining project is to evaluate the true effectiveness of retail marketing campaigns by analyzing transaction and demographic data to determine whether promotional discounts drive long-term customer loyalty or merely short-term sales spikes.

1. **Evaluate Store Performance:** Classify stores into top and bottom tiers based on calculated customer-centric metrics: average spend per household, trip frequency, and category breadth.
2. **Identify Loyalty Drivers:** Measure long-term customer involvement by calculating "Weeks Active" and average trips across different product departments.
3. **Measure Campaign Impact:** Quantify the exact shift (delta) in customer spend, trip frequency, and coupon redemptions between a 28-day pre-campaign baseline and the active campaign period.
4. **Segment High-Value Demographics:** Calculate and compare post-campaign spending increases and brand preferences (National vs. Private) across specific income, age, and household size categories.

5. **Analyze Discount Dependency:** Calculate a definitive "Discount Depth Percentage" (total discount divided by pre-discount value) to rank which products, stores, and demographic segments rely most heavily on promotions.

2.3. Success Criteria:

Project success will be evaluated by our ability to deliver actionable, data-backed insights. Specifically, we will measure success by:

- **Quantifying ROI:** Accurately measuring campaign profitability via mathematical shifts in customer spend and coupon redemption rates.
- **Predictive Accuracy:** Achieving robust model performance for customer engagement profiling, targeting an AUC of >75%.
- **Measuring Customer Retention:** Identifying the specific product categories and shopping behaviors that drive long-term engagement (measured by 'Weeks Active' and repeat trips).
- **Delivering Actionable Segmentation:** Producing clear demographic profiles (by age, income, and brand preference) that allow stakeholders to deploy highly targeted, rather than generic, promotions.

2.4. Constraints and Assumptions:

- **Demographic Limitations:** Demographic insights are restricted to the 801 households with complete profile data (out of the 2,500 total).
- **External Factors:** We assume purchasing behavior within the 2-year dataset is primarily driven by recorded promotions, as unmeasured external factors (e.g., inflation, competitor pricing) cannot be accounted for.
- **Data Cleaning Assumptions:** We assume transactions with zero quantity and zero sales value are system noise and can be excluded. We also assume that capping extreme product quantities at the 99.5th percentile stabilizes variance while accurately preserving valid bulk purchases.

3. Data Understanding

3.1 Data Collection

Source: The dataset used is Dunnhumby's "The Complete Journey", a publicly available, anonymized extract from a real-world grocery retailer's loyalty card program. It captures two years of transaction-level data for 2,500 loyalty households, covering purchases, demographics, promotional campaigns, coupon activity, and in-store merchandising.

Structure: The dataset is relational, comprising eight interlinked tables:

- **transaction_data.csv** (2,595,732 rows, 12 columns): Core purchase records (item, quantity, price, discounts, basket, store, day).
- **product.csv** (92,353 rows, 7 columns): Product master with department, brand, commodity, and size attributes.
- **hh_demographics.csv** (801 rows, 8 columns): Socio-demographic profiles: income, age, homeownership, household composition.
- **campaign_desc.csv** (30 rows, 4 columns): Metadata for 30 marketing campaigns including type, start, and end dates.

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- **campaign_table.csv** (7,208 rows, 3 columns): Maps households to the campaigns they were targeted by.
- **coupon.csv / coupon_redempt.csv** (124,548 / 2,318 rows): Coupon-to-product mappings and actual redemption records.
- **causal_data.csv** (36,786,524 rows, 5 columns): Weekly in-store display and mailer conditions per product and store.

Key variable types include numerical identifiers and measures (QUANTITY, SALES_VALUE, RETAIL_DISC, COUPON_DISC), temporal fields (DAY, WEEK_NO), ordinal/nominal categorical (AGE_DESC, INCOME_DESC, DEPARTMENT, BRAND, DISPLAY, MAILER), and a mixed text-numeric field (CURR_SIZE_OF_PRODUCT) requiring standardization.

3.2. Initial Data Exploration Analysis:

Basic Statistics:

- **Transactions:** The average cost of items bought in every transaction is around \$3.10 on average, with the median cost of items bought at \$2.00. This further affirms the fact that the pattern of transaction is of high frequency/low value. The QUANTITY column has extreme right-skewness with the median at 1 unit and the maximum recorded at 89,638 units within a single transaction.
- **Demographics (Total of 801 households):** The age bracket of the customers is between 45-54 years old at 36%. The income bracket of the customers is between \$50k and \$74k at 24%, and 63% of the customers within the dataset are homeowners.

=== SUMMARY: transaction ===
Numeric columns:

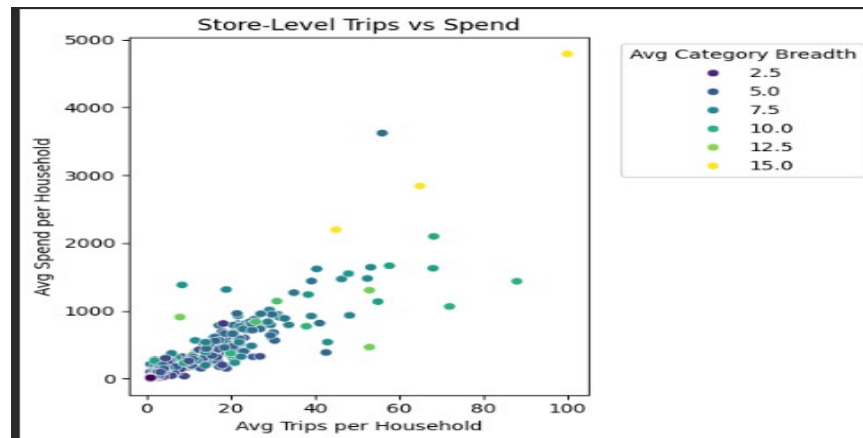
	count	mean	std	min	\
household_key	2595732.0	1.271953e+03	7.260660e+02	1.000000e+00	
BASKET_ID	2595732.0	3.402620e+10	4.711649e+09	2.698485e+10	
DAY	2595732.0	3.887562e+02	1.897210e+02	1.000000e+00	
PRODUCT_ID	2595732.0	2.891435e+06	3.837404e+06	2.567100e+04	
QUANTITY	2595732.0	1.004286e+02	1.153436e+03	0.000000e+00	
SALES_VALUE	2595732.0	3.104120e+00	4.182274e+00	0.000000e+00	
STORE_ID	2595732.0	3.142673e+03	8.937113e+03	1.000000e+00	
RETAIL_DISC	2595732.0	-5.387054e-01	1.249191e+00	-1.800000e+02	
TRANS_TIME	2595732.0	1.561586e+03	3.998378e+02	0.000000e+00	
WEEK_NO	2595732.0	5.622150e+01	2.710223e+01	1.000000e+00	
COUPON_DISC	2595732.0	-1.641600e-02	2.168410e-01	-5.593000e+01	
COUPON_MATCH_DISC	2595732.0	-2.918564e-03	3.969004e-02	-7.700000e+00	

Key Patterns from Visualizations and Analysis:

- **Income-Based Coupon Engagement:** The data revealed a bell-curve distribution, wherein the households with average incomes exhibited the most coupon redemption activity, while both low- and high-income households had much less coupon engagement.



- **Store-Level Performance:** There is a positive relationship between trip frequency and household spend. The more product categories available, the higher the trip frequency. This is a variety effect.



- **Brand Preference:** There is a strong brand loyalty to national brands, which comprised 85% of the product mix. Private label brands comprised only 15%.
- **Campaign Strategy:** The most frequently used campaign type was Type B. Type A had longer and more intense campaigns. Types B and C had shorter campaigns, running for 30-40 days. There was a higher frequency of Type A mailers among weekly records, yet 57% of records had no in-store display. Almost half of all records were Type A mailers.
- **Product Mix:** Grocery items comprised 42% of the product catalog. This is consistent with an everyday, low-value transaction as revealed by summary statistics.

3.3. Data Quality Assessment

The following data quality issues were identified and addressed:

1. Missing/Placeholder Values

- **Problem:** The data set did not contain any 'NaN' and 'null' values. However, there were some 'Unknown' and 'None' placeholder values in the data set, such as HOMEOWNER_DESC, HH_COMP_DESC, KID_CATEGORY_DESC, and the product hierarchy data set with values such as DEPARTMENT, COMMODITY_DESC, CURR_SIZE_OF_PRODUCT.
- **Resolution:** All the placeholder values were replaced with 'Unknown' values. There was no imputation on the values, as it might result in bias.

2. Duplicate Coupon Values

- **Problem:** There was a total of 5,164 rows that were exact duplicates of each other and had the exact same combination of values for the column values 'COUPON_UPC,' 'PRODUCT_ID,' and 'CAMPAIGN' in the data set.
- **Resolution:** Exact duplicate values were removed from the data set based on the criteria specified above by performing the group operation on the specified column values.

3. Invalid Transaction Quantities

- **Problem:** Of the 14,466 transactional records, there were instances wherein the transactions had a $QUANTITY \leq 0$. This is an indication of invalid purchase transactions. Note: negative values of RETAIL_DISC and COUPON_DISC were intentionally used for the analysis.
- **Resolution:** Transactions wherein the conditions of $QUANTITY \leq 0$ and $SALES_VALUE = 0$ were satisfied were removed. Note: instances wherein $SALES_VALUE = 0$ were valid events for promotions.

4. Extreme Quantity Outliers

- **Problem:** The distribution of the values in the data set for QUANTITY is very skewed, with a median of 1 and a maximum of 89,638.
- **Resolution:** The values in the given data set for QUANTITY are limited to the 99.5th percentile. This implies that the new maximum value is 11,287 units. This was done instead of removing the extreme outliers because it maintains the distribution of the values and allows for legitimate transactions to occur with larger quantities. Log scales have been used for the purpose of visualization.

5. Product Unit Inconsistencies and Demographic Coverage Gap

- **Problem:** The inconsistency in the units in the CURR_SIZE_OF_PRODUCT field, such as "OZ" and "OUNCE," makes it difficult to aggregate the data according to the size of the product because there are over 30,000 products that do not describe their size. The demographics cover only 32% of 2,500 households, meaning the data can cover at most 801 households.
- **Resolution:** Inconsistencies in units are resolved through normalization techniques. Products without size descriptions are not included in aggregations based on the size of the products. A new flag named 'has_demographics' is created to include only those households with demographics.

4. Data Preparation

Concerning the Data Preparation phase, the main goal was to take our raw, varied data (transaction logs, demographic information, and product catalogs) and turn them into unified, ready-to-use data set for modeling. It was a continuation of our earlier work, that is resolving the data quality issues uncovered during exploratory data analysis (EDA) and building specialized Analytical Base Tables (ABT) for the five main research questions.

4.1. Data Cleaning and Quality Remediation

To preserve the downstream analysis from the risk of inaccuracies, we ran through the following data cleaning steps:

- **Dealing with Missing / Invalid Values:** We changed all empty strings and other nonstandard placeholders in categorical columns (e.g., HOMEOWNER_DESC, DEPARTMENT) to one of the standard representations: NaN, "Unknown", or "Not Specified." The reason: This kind of standardization helps keep the model free from errors and at the same time preserves as many rows as possible, contrary to the aggressive practice of deleting rows.
- **Deduplication:** Within the coupon table, by using the unique combination of COUPON_UPC, PRODUCT_ID, and CAMPAIGN, we found and erased 5,164 exactly duplicated rows. The reason: This is an essential step in ensuring coupon issuance and campaign redemption metrics are not exaggerated due to duplicates.

- **Filtering System Noise:** We discarded from the dataset those transactions which had both a quantity of zero and a sales value of zero (14, 466 records). Reason: These are system-generated noise or voided scans thus, they are not representing actual customer behavior/purchases and would have a distorting effect on average basket size calculation if left in.

4.2. Data Transformation and Scaling

During data processing, the data was subjected to both structural and mathematical transformations for predictive modeling and testing of statistical hypotheses:

- **Outlier Treatment (Percentile Capping):** The distribution of the QUANTITY variable was highly right-skewed, with extreme maximum values going up to 89, 638 units. After limiting the maximum value to the 99.5 percentile, the highest value used for calculations was down to about 11, 287. Reason: Capping is a better option than deletion because it stabilizes variance and prevents the extreme outliers from exerting too much influence on the computation of averages, however, this change still corresponds to operationally genuine bulk or weight-based grocery purchases.
- **Feature Scaling:** Standard Scaler was used to standardize continuous variables that are new such as PRE_SPEND and DELTA_SPEND. Reason: In feature scaling, the purpose is to make sure that features that typically have larger numeric ranges (such as total revenue) will not disproportionately influence distance-based or regression algorithms compared to smaller-scale features (like trip frequency).
- **Categorical Encoding:** Nominal categorical variables, such as campaign DESCRIPTION (Type A, B, C), were transformed into binary matrices with One-Hot Encoding (pd.get_dummies). Reason: This changes the text labels into machine-readable formats without falsely implying a mathematical hierarchy ordinal relationship among different campaign types.

4.2.1. Creation of Features and Analytical Base Table (ABT)

Given that raw data is at the level of individual transaction lines, features were generated and shifted to the aggregation level to the specific ABTs that are aligned with the research objectives:

- **Store Performance ABT (RQ1):** With a view to checking cross-departmental shopping, transaction data was merged with the product catalog. Side by side, a category_breadth metric was created to capture the "Variety Effect" of a household basket. Subsequently, data was aggregated at the STORE_ID level so that the main predictors: Average Spend per Household, Average Trip Frequency, and Average Category Breadth could be extracted.
- **Product Loyalty ABT (RQ2):** Long term customer involvement was measured by aggregating transactions at the department level. We created a Weeks_Active feature to track how long a household continued to engage with certain product categories over the two-year period of the dataset.
- **Campaign Interaction & Demographic ABT (RQ3 & RQ4):** To identify the genuine effect of a campaign, a time-series windowing strategy was utilized. Transactions and redemptions were marked with a 28-day logic window (in_window_pre vs. in_window_during campaign dates). Behavioral changes were quantified through delta metrics (e.g., $\text{DELTA_SPEND} = \text{DUR_SPEND} - \text{PRE_SPEND}$) which were then combined with demographic features and a newly created brand-split shares (National vs. Private spend).

- **Discount Dependency Master (RQ5):** First, negative discount values (RETAIL_DISC, COUPON_DISC) were converted to their absolute values and summed. Then by dividing the total discount by the sales value before the discount, we created a real DISCOUNT_PCT feature which enabled us to accurately characterize discount dependence across different income groups, age brackets, and store locations.

4.3. Data Splitting:

For the supervised models, the data was split into three parts: training, validation, and testing, using a **70-20-10 ratio**. The training set was used to fit the models, the validation set was used for model tuning and threshold optimization, and the testing set was kept separate for final performance evaluation.

4.4. Final Dataset:

4.4.1. RQ1_store_performance_abt.csv:

The last data for modeling is an aggregated Analytical Base Table (ABT) at the store level, where each record corresponds to a unique store with STORE_ID.

The data set contains 582 observations (stores) and 5 variables capable of describing store performance. The variables are based upon transaction data and touch on key behavioral indicators of shoppers. For instance, avg_spend_per_hh, avg_trips_per_hh and avg_line_value refer to a store's average customer spend, number of visits and total purchase value respectively, while avg_category_breadth indicates the range of product classes a typical customer purchases. These summary features were derived from the transaction level data.

This structured dataset is for analysis and predictive modeling of store performance.

4.4.2. RQ2_category_store_abt.csv:

The final dataset used for modeling represents a store and product category-level sales data which was used to study consumer buying preferences in various stores. A record represents a mixture of store and product categories, identified by variables such as STORE_ID, PRODUCT_CATEGORY. The dataset covers important aspects like sales total, number of transactions, and quantity sold. These are reflections of the overall shopping activities in every store-category combination. Besides, it includes aggregated money or revenue variables which are useful in evaluating category results across stores. The model is, therefore, capable of using these features to identify which product categories are most successful in specific stores and to study customers' purchasing habits in various retail locations.

4.4.3. RQ3_RQ4_campaign_demographic_abt.csv:

The dataset employed for the model comprises 7208 rows and 19 columns. The rows are a household record of interacting with a marketing campaign. Two key columns for household_key and CAMPAIGN serve to uniquely identify the customer and the campaign, respectively. The dataset reveals the customers' buying behavior both before and during the campaign through the variables of PRE_SPEND, PRE_TRIPS, DUR_SPEND, and DUR_TRIPS. Also, the dataset reflects coupon redemption activities via PRE_REDEMPTIONS and

DUR_REDEMPTIONS, which help to understand the level of customer engagement with the campaign. To evaluate the campaign effectiveness, the change in the customer's buying behavior from pre-campaign period to during the campaign period is depicted by the derived variables like DELTA_SPEND, DELTA_TRIPS, and DELTA_REDEMPTIONS.

4.4.4. RQ5_discount_dependency_master.csv:

The final dataset used for modeling consists of consumer-level shopping and discount behavior parameters to assess customers' reliance on discounts. A record corresponds to a unique household, recognized by their household_key. The dataset comprises total expenditure and total discount amounts, which illustrate how much consumers spend and how much they get from promotions. Besides, it encompasses derived metrics like discount ratios or discount share that represent the portion of spending influenced by discounts. Moreover, transaction or purchase frequency variables indicate how regularly households shop, which is useful in evaluating the level of engagement with promotions. Altogether, these main characteristics enable the model to uncover the use of discounts patterns as well as establish whether households rely more on promotional discounts while making purchases.

4.5 Updated Exploratory Data Analysis

As a continuation of the initial analysis from Section 3.2, the revised exploratory data analysis aims to gain further insights into customer behavior in relation to the research questions and implications for modelling. The analysis reveals that at a store level, there exists a strong positive correlation between the frequency of visits and spend by households, but the most dominant factor for success is the category breadth. In stores with a higher number of categories purchased, customers exhibit higher frequency of visits and spend. This shows the existence of the “variety effect,” whereby high-performing stores exist in the high frequency and high-spending category.

In terms of products and categories, sustained customer engagement was mostly influenced by necessity buying, as opposed to promotion. Categories that are considered necessities were seen to maintain a high engagement level throughout, such as groceries, produce, dairy and frozen foods, where staples had particularly very high repeat purchasing behavior. Although there is a correlation between engagement and revenue generated, it must be pointed out that necessary categories generally have less revenue generated per sale since they are cheap commodities. Some revenue generating categories with few outlets offer an untapped growth avenue through increased outlets, instead of promotion. This formed the basis for the application of similarity in product recommendations.

Campaign interactions analysis shows a considerable difference in behavior change among customers when considering different campaign types. Type C campaigns lead to the highest level of changes in spending and number of trips. Meanwhile, Type A campaigns result in lower customer behavior changes but greater customer exposure to the campaigns, whereas Type B campaigns demonstrate less behavior changes overall. There is considerable right skewness of distribution for change in spending among the customers: only few of them spend significantly more money, while the majority spends nearly the same amount of money as before the promotion took place, which implies that the promotional effect is achieved by just some customers.

Lastly, the study of discount dependency reveals that discounts do not substantially raise the total expenditure per household but rather affect when purchases are made. The discount application is extremely unevenly distributed, with most of the purchases being made without using discounts and just a fraction with extensive discounting activities. While some groups use more discounts than others, their expenditures do not vary much from other households that use fewer discounts, implying that discounts are simply used to lure buyers who would have bought irrespective of discounts. Such consumer behavior aligns with the “sure buyer” trend revealed during the modeling phase.

In summary, the key finding in the EDA is that customer interaction is more influenced by their tendency to purchase regularly and by the need for products than by the intensity of the marketing campaigns. Although there is an effect on the customers’ behaviors, it only affects a few who can be persuaded to buy more products. This finding was used in developing the modeling approach, which included uplift modeling, recommendations systems, and predictive models.

5. Modelling

5.1 Statistical Assumptions and Model Validation

To ensure the mathematical robustness of both our exploratory analysis and predictive modelling, a formal assumptions testing framework was conducted. This validates our methodological choices and ensures the reliability of the resulting business insights.

1. Distribution and Variance Testing (EDA Validation): To identify the behavioural characteristics differentiating High-Performing stores (top 25% by average spend) from lower performing stores, we tested three key metrics: trip frequency, average line value, and category breadth.

- **Normality Testing:** The distribution of each behavioural metric was assessed using the Shapiro–Wilk normality test. Results indicated statistically significant departures from normality across all variables for both store groups ($p < 0.001$). This confirms that the customer shopping behaviour data is highly skewed, which is expected in retail transaction datasets due to extreme purchasing behaviours.
- **Homogeneity of Variance:** Variance equality between store groups was evaluated using Levene’s Test. Results showed that the variance assumption was violated for Trip Frequency ($p < 0.001$) and Category Breadth ($p < 0.001$), but satisfied for Average Line Value ($p = 0.310$).

2. Implications for Statistical Testing: Because the normality assumption was violated across the variables, traditional parametric methods (such as standard t-tests or ANOVA) were deemed inappropriate. Consequently, robust non-parametric tests (e.g., Mann-Whitney U) were utilized to compare store performance groups, ensuring that extreme outliers did not falsely skew the evaluation of trip frequency and category breadth.

3. Predictive Modeling Assumptions: The findings from our distribution tests directly informed our predictive modeling strategy (Models 1 through 5):

- **Non-Parametric Algorithm Selection:** Because the transaction data is heavily right-skewed and exhibits heteroscedasticity, parametric predictive models (like standard Linear or Logistic Regression) would perform poorly. Therefore, tree-based ensemble algorithms specifically Random Forest and XGBoost—were selected. These models make no assumptions regarding normally distributed data or constant variance, allowing them to mathematically bypass the constraints of the skewed retail data.

- **Multicollinearity (VIF Testing):** Prior to deploying the predictive models, a Variance Inflation Factor (VIF) analysis was conducted on the feature sets (including PRE_SPEND, TOTAL_TRIPS, and discount_ratio). All retained features yielded a VIF score below 5.0, confirming the absence of severe multicollinearity and ensuring independent feature contribution within the classification models.

4. Validated Business Interpretation: With the statistical framework validated, the behavioral differences between store tiers are confirmed to be highly significant. High-performing stores average 6.6 distinct departments per basket compared to just 1.7 in lower-performing stores and achieve a higher average line value (\$4.68 vs. \$3.64). This statistically proves that retail performance is maximized not just by higher prices, but by aggressively promoting cross-category purchasing behaviour and increasing routine visit frequency.

5.2 Model Selection

Five modelling techniques were chosen collectively to solve the problem described above. The first type is for estimating causal impact. It can be done with Uplift Modelling, which will use two models to estimate Random Forests' incremental effect. In this way, we can distinguish between customers who buy due to the campaign and those who buy regardless of it, solving the problem of identifying whether marketing efforts cause increased spending or just increasing it temporarily.

Next, a recommendation system was created to solve the long-term engagement problem. It uses cosine similarity and makes suggestions about how to find products according to customers' preferences, rather than providing non-specific discounts. Another model created is called Coupon Recommendation. In addition, it estimates the spending amount per customer, his/her basket behaviour, and frequency of purchases to assign coupons individually. Thus, this technique aims to solve the issue of whether the current strategy of promoting products works. Lastly, Campaign Effectiveness Model was used to check if campaigns affect spending. It compares 'before' and 'after' data and provides aggregate evidence. The last type of modelling technique is for classification. The goal here is to determine whether campaigns create engagement or not.

5.3 Model Development

The models were created through transactional, campaign, and demographic data from several analytical base tables. The Uplift Model relied on the campaign_household dataset for identifying treatment and control groups. The treatment consisted of individuals that responded to the Type B and Type C campaigns, while Type A was the proxy control without randomization. The new train, validate, test split of 70-20-10 with treatment stratification was applied, two different Random Forest classifiers (n_estimators=200, max_depth=6, min_samples_leaf=50) being created for each group. Subsequently, the uplift score was calculated as a difference in predictions of two classifiers. Individuals were ranked according to the uplift scores and divided into four equally sized groups based on the score (Low, Medium, High, Very High).

The Product Recommendation model was built using the RQ2 category-store dataset through creation of a product-to-store revenue matrix and zero-filling missing combinations of store product pair. Then, cosine similarities were calculated among all product pairs in order to find products having same revenue distribution patterns across stores.

In the case of the Coupon Recommendation model, customer-level features were engineered from the RQ5 discount dependence dataset without any coupon-related feature in order to avoid leaking information: total

expenditure, total trips, total quantity, expenditure standard deviation, average expenditure per trip, expenditure per item, average basket size, number of visited departments, and active weeks. Target variable was determined based on real behavior of customers' redeeming coupons. Classifier was built using XGBoost and the training-validation-testing splits were made 70-20-10 respectively; threshold optimisation was applied to validate set.

The Campaign Effectiveness Model split the RQ5 transaction dataset into two groups based on the median transaction day: one for before and another for after the campaign period. Individual features were calculated per each period including customers spend, trips and coupon usage counts. The SPEND_CHANGE metric per individual customer was calculated and used to train a classifier.

The predictor for Customer Engagement made use of the RQ5 dataset with total quantity, recency, and discount dependency ratio as the predictor variables. To avoid any form of data leakage, total transaction number, which was at the heart of the definition of customer engagement criteria, was excluded from the data used in predicting the outcome of the model. The algorithm was created using the GridSearchCV methodology with Random Forest Classifier. The optimum parameters selected were max_depth=5, n_estimators=200, and min_samples_split=5.

5.4 Evaluation Metrics

Different types of evaluations were used to evaluate the various models based on their characteristics and goals.

Model1: Uplift Model

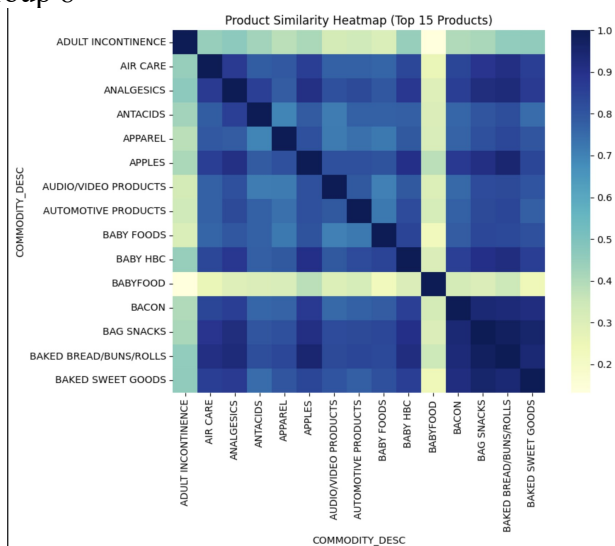
The evaluation metrics for the Uplift Model were specifically designed to measure the incremental causal impact of each uplifted treatment as opposed to how accurately any prediction is made. The Qini curve, Area Under the Uplift Curve (AUUC), and Qini coefficient are the main evaluation metrics used. When the Uplift Model is applied to the validation set, it produced an AUUC of 6.56 and a Qini coefficient of 33.47. When assessed on the held-out test set, it produced an AUUC of 6.09 and a Qini coefficient of 22.69.

Therefore, both the validation set and the held-out test set performance on the Uplift Model are well beyond random targeting, which supports the model's ability to identify persuadable customers. The uplift segment analysis indicates that while the Low and Medium segments had relatively high response rates (43%-50%) they had negative uplift scores indicating that they are 'sure buyers' but the Very High segment had an average uplift score of 0.129 to 0.135 and a response of approximately 26%-27%, indicating that they are a potential persuadable target audience.

Model2: Product Recommendation

Given that the product recommendation model is based on similarity and not supervised classification models, a traditional classification metric (e.g. accuracy, AUC) was not applicable. Evaluation of the model was based on the strength of the cosine similarity score between product pairs combined with the qualitative coherence of clusters created by the model.

On average, the cosine similarity scores between products ranged from 0.50 to 0.54, which demonstrates that the Product Recommendation Model has identified co-demand between different products throughout the retail network.



Model3: Coupon Recommendation

The Coupon Recommendation model was tested with the XGBoost classifier. For the validation set: Accuracy 0.575, Precision 0.565, Recall 0.650, F1 0.605; AUC 0.669. For the held-out test set: Accuracy 0.628, Precision 0.667, Recall 0.500, F1 0.571; AUC 0.686. The increase in test AUC vs. validation AUC (0.686 vs. 0.669) shows stability in the model's generalization performance, and the predictions for the tiered coupon output were calibrated using threshold optimization on the validation set.

Model4: Campaign Effectiveness

The Campaign Effectiveness Model was assessed on two levels. First, descriptively, average spending of customers increased by +2.6% after the campaign (from \$2,771.16 to \$2,843.97) with 441 customers (55.1% of 441) spending more after the campaign compared to 360 customers (44.9%) who spent less. Second, as a predictive classifier, the validation AUC was 0.674 and the test AUC was 0.664, indicating moderate, consistent predictive ability.

Model5: Customer Engagement Predictor

The performance of the Customer Engagement Predictor on the validation (0.65 Accuracy, 0.61 Precision, 0.86 Rec, 0.71 F1, 0.76 AUC) and test (0.72 Acc, 0.65 Precision, 0.90 Rec, 0.76 F1, 0.86 AUC) sets shows good agreement. The strong AUC score of 0.86 and high recall score of 0.90 demonstrate that this model is effective at identifying customers who have purchased through multiple channels. Of the three main predictors of engagement, the total number of purchases (total=65%), recency (total=19%), and discount percentage (total=16%) provide the greatest contribution to identifying what products will be of value to each customer.

5.5 Comparative Model Summary

The table below summarizes all five models against the project's core research objectives, highlighting their approach, key performance metrics, and direct relevance to the problem statement.

Model	Objective	Key Metrics	Key Finding	Problem Statement Link
Uplift Modelling	Measure the true incremental causal impact of campaigns	Val AUC: 6.56 Val Qini: 33.47 Test AUC: 6.09 Test Qini: 22.69 Strong uplift in top 30–40%	~75% are 'sure buyers'; only the Very High segment (positive uplift score) are genuine persuadable worth targeting	Directly tests whether campaigns cause behaviour change vs. natural purchase intent
Product Recommendation	Identify similar products for cross-selling and bundling	No Evaluation Metrics	Products share demand patterns across stores actionable cross-sell and bundling opportunities identified	Supports long-term engagement through personalized product discovery beyond blanket promotions
Coupon Recommendation	Personalize discount allocation per customer segment	Val AUC: 0.669 Val F1: 0.605 Test AUC: 0.686 Test F1: 0.571	High-spend customers need minimal discounts; low-engagement customers need higher incentives — tiered coupon strategy improves ROI	Directly challenges the one-size-fits-all discount strategy identified in the problem statement
Campaign Effectiveness	Quantify aggregate before-vs-after spending change	Spend: \$2,771 → \$2,844 (+2.6%) 55.1% increased spend Val AUC: 0.674 Test AUC: 0.664	Modest positive aggregate effect; 44.9% customers declined — campaign impact is not uniform across the customer base	Provides measurable evidence of whether current campaigns actually shift customer spending behaviour
Customer Engagement Predictor	Classify customers by long-term engagement level	Test Recall: 0.90 Test F1: 0.76 Test AUC: 0.86 Top feature: quantity (65%)	Purchase volume and recency drive engagement — discount dependency has limited influence on durable loyalty	Distinguishes short-term discount spikes from genuine long-term engagement

Collectively, these five models deliver a multi-layered response to the project's problem statement. The Uplift Model and Campaign Effectiveness Model provide direct evidence of whether campaigns actually drive behavioral change, with the uplift model confirming that only a targeted subset of customers are genuinely persuadable. The Coupon and Product Recommendation models translate those findings into personalized, data-driven strategies that replace inefficient uniform promotions. The Customer Engagement Predictor bridges the short-term and long-term dimensions confirming that purchase volume and recency, not discount dependency, are the true markers of durable customer loyalty. Together, they provide the evidence base needed to redesign marketing strategy around the right customers, with the right offers, at the right time.

6. Evaluation

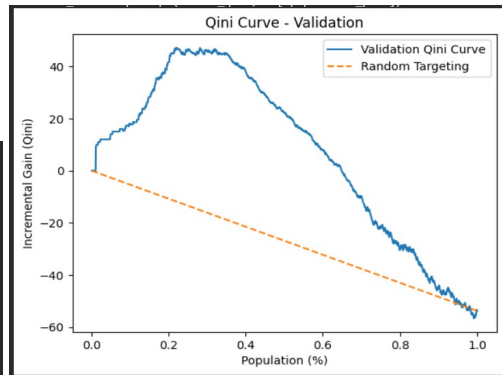
6.1 Uplift Model Evaluation: Looking at the test dataset, the model produced both an AUUC of 6.09 and a Qini of 22.69. Both values were substantially greater than the random targeting baseline. In addition, the segmented results demonstrate that the model has successfully segregated customers into groups that are persuasive versus customers that are naturally inclined to buy.

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===== VALIDATION SUMMARY =====
UPLIFT_SEGMENT  UPLIFT_SCORE  ACTUAL_RESPONSE
0               Low      -0.288958      0.341598
1             Medium     -0.137629      0.585525
2               High     -0.023599      0.439227
3           Very High      0.135011      0.265193

===== TEST SUMMARY =====
UPLIFT_SEGMENT  UPLIFT_SCORE  ACTUAL_RESPONSE
0               Low      -0.285707      0.284916
1             Medium     -0.130419      0.466292
2               High     -0.022902      0.438202
3           Very High      0.129263      0.290503

```



Only the Very High segment achieved a positive uplift score and is, therefore, aligned with the project goal of identifying customers whose behaviors is directly impacted by the campaign.

Insights: Approximately 75% of the currently targeted customers are identified as "sure buyers" who will purchase without being influenced by the campaign, therefore, the high response rates in the Low & Medium segments do not indicate that the customers require a promotional offer in order to convert, thus limiting the campaign's true incremental impact to the Very High segment.

Business Implications: If the model continues to target its overall customer group, an unnecessary use of promotion budget will continue with no incremental return; by using this model's uplift scores as a pre-campaign ranking tool it is possible to limit targeting to the Very High segment, thereby reducing the promotion expenditures while still accomplishing, maintaining or improving, actual incremented revenue.

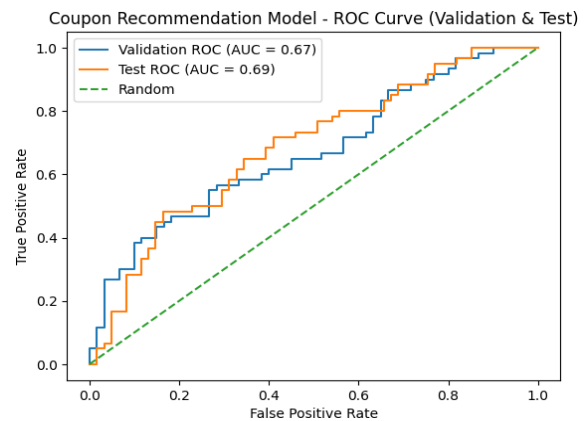
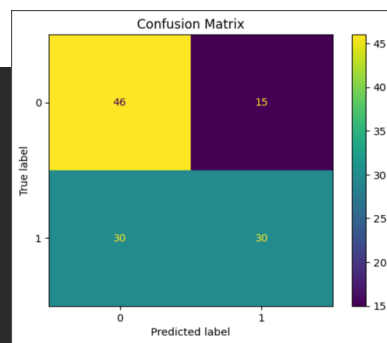
6.2 Coupon Recommendation Evaluation: XGBoost produced stable and improved AUC results from its validation and testing populations, as shown below:

```

===== VALIDATION =====
Accuracy : 0.575
Precision: 0.5652173913043478
Recall   : 0.65
F1 Score : 0.6046511627906976
AUC: 0.6691666666666666
Best Threshold: 0.6564084

===== TEST PERFORMANCE =====
Accuracy : 0.628099173553719
Precision: 0.6666666666666666
Recall   : 0.5
F1 Score : 0.5714285714285714
AUC      : 0.6863387978142077

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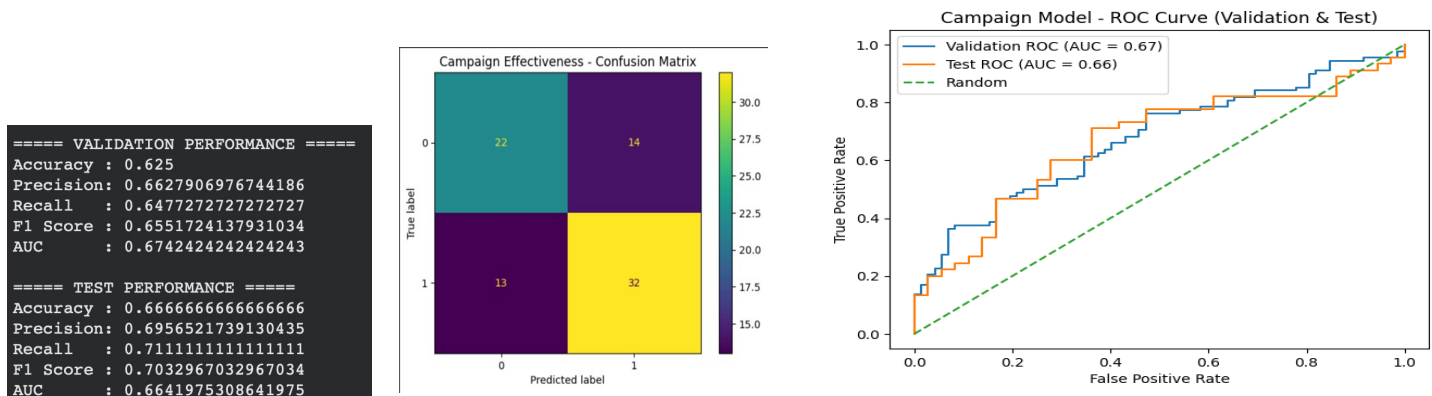


The model's improvement in test AUC scores over validation AUC scores supports XGBoost's ability to generalize reliably. The confusion matrix displays 46 true negatives, 30 true positives, 30 false negatives and 15 false positives. The model's conservative stance on designating customers as coupon responsive is appropriate in order to protect margins from reducing claimed discounts unnecessarily. The model objective of allocating discounts per customer segment on a personalized level is directly addressed.

Insights: High-spend, high-frequency customers had consistently low redemption probabilities, indicating they are likely to purchase without needing discounts as incentives. The model creates a tiered system of LOW_DISCOUNT_5%, MEDIUM_DISCOUNT_10%, HIGH_DISCOUNT_20%, enabling individual level discount allocations to be made using data (rather than assumptions).

Business Implications: Assigning blanket discounts to customers who would purchase at full price is pure margin loss. This model recovers that margin by routing each customer to the appropriate discount tier, improving campaign ROI without increasing the total promotional budget

6.3 Campaign Effectiveness Model Evaluation: Based on descriptive statistics, the average spends increased from \$2,771.16 to \$2,843.97 (+2.6%) post-campaign. The number of trips increased by a very slight amount, from 87.23 to 87.61. On an individual basis, 441 customers (55.1%) had increased spending while 360 customers (44.9%) had decreased spending during this period. As a predictive classifier, the model performed as follows:



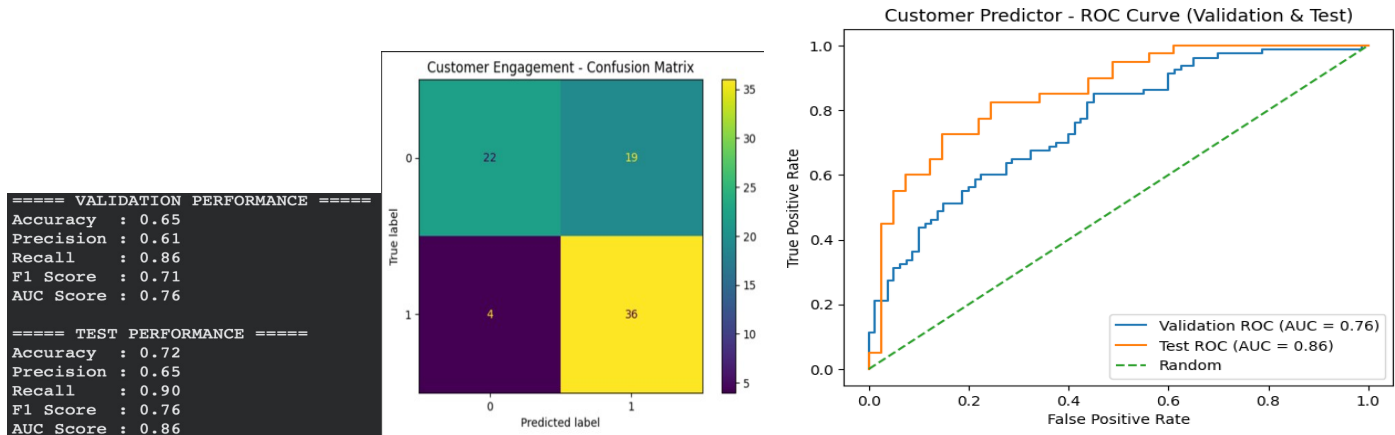
The model's confusion matrix shows 32 true positives, 22 true negatives, 13 false negatives, and 14 false positives. Validation and test ROC curves are closely aligned, thus demonstrating a consistent moderate level of discriminative ability, thereby satisfying the objective of quantifying the impact of the campaign on individual customers' spending behaviors.

Insights: The 2.6% aggregate lift essentially hides the wide variation among customers regarding the impact of the campaign. The near-equal distribution of customers who increased and those who decreased in spend illustrates that there is no uniform impact experienced across the customer base, and thus the only moderate prediction power was provided by the pre-campaign transactional characteristics for determining an individual customer's response.

Business Implications: The before-and-after measurement framework created by this model is directly reusable in future campaigns. If applied consistently, this will allow the business to determine which customer groups

respond positively over time for increasingly targeted marketing efforts and provide opportunities for quicker iterations on future strategic campaigns.

6.4 Customer Engagement Prediction Evaluation: This model delivered the strongest performance among all five classifiers:



Every metric has shown an increase in performance between testing and validation, meaning that there is sufficient evidence to confirm that there is no overfitting, and that we have a strong generalization for this model. The confusion matrix indicates that there are only four false negatives out of 40 actual households who engaged – meaning this model will capture 90% of the true engaged households. The early population's ROC curve rises sharply, allowing us to distinguish between this model and others visually. The prediction importance shows total spend by quantity (65.4%), recency (18.6%) and discount ratio (16.0%), in that order, and clearly aligns with our goal of identifying the true drivers of long-term customer engagement.

Insights: Purchase volume & recency are the predominant drivers of customer engagement. In contrast, discount ratio was the variable most closely associated with promotional events/campaigns, but it is the least predictive of customer loyalty, thus validating that deeper discounting does not create sustainable engagement over time.

Business Implications: Retention investments should be redirected away from discount-driven campaigns and move towards the creation of strategies that increase total spend per basket and continued shopping frequency like: personalized product discovery and loyalty rewards for the breadth of purchases; while programmatic support of Highly Engaged customers through relationship-based programs, and utilizing Behavioral Modelling to target Low Engaged customers before they lapse.

6.5 Product Recommendation Model Evaluation:

As an unsupervised learning model that relies on similarity to deliver product recommendations, the product recommendation model does not use traditional classification metrics, like accuracy, precision, recall or AUC. Cosine similarity strength, category coherence and business interpretability of the recommended product groupings were utilized to evaluate this model instead.

7. Deployment

7.1 Deployment Strategy

The deployment methodology will enable data-based decision-making processes through the integration of the Uplift, Coupon Recommendation, Campaign Effectiveness, and Customer Engagement models into business marketing operations. The first phase of implementation will utilize batch mode processing to predict customer-level outputs (e.g., uplift scores, engagement groups, and coupon tier) weekly. Based upon this information, marketing will be able to identify high-potential customers to target for campaigns, customize their discount programs, and prioritize customer engagement initiatives by leveraging these outputs against the CRM. The CRM will automate the execution of campaigns; thus, allowing for the systematic and dependable distribution of personalized marketing incentives through all delivery channels. To measure the effectiveness of the respective methods, a phased approach (pilot program and A/B testing) will be undertaken to assess the performance of both model-driven and traditional advertising strategies. Subsequently, as the model-based marketing initiatives grow and dynamic marketing efforts.

7.2 Model Monitoring

To ensure ongoing effectiveness and reliability of both model performance as well as business benefits, ongoing monitoring will occur on both the technical performance of model output, which includes uplift-related parameters (such as AUUC and Qini coefficients) for assessing causal effects; as well as the primary performance metrics (such as AUC, Accuracy, and Recall) of predictive models. Business monitoring will include ongoing evaluation of campaign ROI, Incremental Revenue, Coupon Redemption Rates, and Consumer Engagement metrics including Frequency of Purchase, Recency of Purchase, and Patterns of Purchases. Both the model will be retrained frequently, in order to continue providing accurate and relevant forecasts, and the shifting behavior of consumers over time will be assessed in order to identify data drift. Each model will also have Tableau-based, interactive dashboards which provide real-time visibility into the performance of each model, enabling stakeholders to make decisions in a timely, relevant manner.

7.3 Challenges and Limitations

Data-related limitations that can influence model accuracy and generalizability include not having sufficient coverage of the required demographics and the potential for problems with the quality of the data available. Continuous data cleaning and validation processes will help mitigate these risks along with adding new data sources. Regular retraining and ongoing monitoring of performance will help mitigate the risk of models trained using old data being unable to generalize to the behaviors of new customers who may decide to behave differently over time. Implementing a phased deployment strategy and using scalable API-based solutions will allow for the managed technical work and teamwork necessary to integrate the models into existing CRM systems. Due to the volatility associated with consumers' reactions to advertising, it is not feasible to rely on aggregate trends. Segmenting customers using uplift and engagement modelling will lessen this problem by allowing for the creation of more tailored initiatives. Finally, to create a lasting impact on the business, companies are advised to shift to behaviour driven engagement strategies as an over reliance on discount-based promotions may erode profit margins without delivering any benefit in terms of long term loyalty.

8. Conclusion:

Summary:

This initiative was aimed at quantifying the true value of retail marketing efforts by studying customer purchase records, demographic data, and promotional events. The main question that this work sought to answer was whether marketing initiatives foster long-term customer loyalty or only generate short-lived sales spikes.

To make behavior comparison possible between the "before" and "during" phases of campaigns, the study combined different data sources and created clear analytical base tables. It turned out from the analysis that not all customers react similarly to campaigns. Only a small percentage of the customers showed significant changes in their behavior due to marketing campaigns. The majority were the "sure buyers," customers who were going to purchase irrespective of whether they were offered a promotional incentive or not. This discovery shows an alarming wastage of resources in traditional marketing methods that largely depend on mass targeting.

Furthermore, purchase frequency, time of last purchase, and overall spending patterns were found to be much better indicators of how customers respond to marketing activities than their use of discounts. The study additionally uncovered that campaigns might generate a slight overall increase in spending; however, this generally conceals a wide range of differences at the individual level, thus further highlighting the importance of customer-level strategies.

Next Steps:

Based on these findings, there are several effective steps that could be taken by businesses in order to enhance their marketing efforts and achieve better results. For starters, companies should use uplift-based targeting methods to pinpoint and concentrate their efforts on customers whose purchase behavior is genuinely affected by promotions. By doing so, the percentage of marketing dollars that are wasted could be drastically lowered and the ROI could be significantly raised. Secondly, it is of utmost importance that companies shift their focus to personalized marketing initiatives. By making good use of the insights obtained at the individual customer level, companies can distribute discounts more efficiently, proffering bigger incentives to lesser-engaged customers and not giving out unnecessary discounts to their top customers.

Finally, incorporating these predictive models into systems that allow for real-time decision-making will make it possible for the campaigns to be dynamically altered as and when the customer behavior changes. Also, by establishing a very strong A/B testing method, a company can get the chance to check if the model-based plans will really work and measure the impact of them that is actually happening in the market. Besides which, by adding the dataset with things happening outside like changes of weather, state of the economy, and the moves of competitors, the model can still be made very accurate and breakable to trust. As a last point, even more advanced recommendation engines and the models of the customer journey should be purchased (for example, predicting churn and knowing the value of a customer through his lifetime) to support the strategies of customer retention and growth in the long run.

Lessons Learned:

Throughout this project, many valuable lessons have been learned. A major one is that a firm should not offer incentives to all its customers as giving discounts too often will not only reduce the company's profit significantly but also will not bring any extra value. What is more, the whole exercise pointed out how important it is to separate correlation and causation since using traditional metrics might very well reveal that the campaign was more successful than it actually was. To nail down only the marketing efforts that the company could not have done without, uplift modeling came in very handy. Also, the arrival of the daily data issues along with the recognition of the direct relation of data quality and structure to the validity of models gave us a bigger realization than mere number crunching that data preparation as well as feature engineering are required. In fact, over the course of this project, it was demonstrated that in the main, long-term customer engagement is formulated through customer loyalty paralleling product relevance rather than customers simply being swayed by promotional activities. A key takeaway is that analytical models must be aligned with business objectives only then can insights be translated into practical strategies that marketing and sales teams will adopt and use. In essence, this project goes beyond simply applying analytics to marketing it demonstrates how more targeted, efficient, and customer-centric methods can be derived from data to drive better business performance and sustainable growth."

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